

Problem Set 3

Demand: Hicks, Slutsky, and Elasticity

EC306: Intermediate Microeconomics

Instructions. Work through these in class. Show all working and interpret your numerical answers in words. Worked solutions follow at the end.

Question 1 — Hicksian demand, expenditure function, Shephard

A consumer has $u(x, y) = x^{0.5}y^{0.5}$.

- (a) Set up the expenditure-minimisation problem (minimise $p_x x + p_y y$ subject to $u = \bar{u}$) and, using the tangency condition $MRS = p_x/p_y$, derive the *Hicksian* (compensated) demands $h_x(p_x, p_y, \bar{u})$ and $h_y(p_x, p_y, \bar{u})$.
- (b) Derive the expenditure function $e(p_x, p_y, \bar{u}) = p_x h_x + p_y h_y$. Verify it is homogeneous of degree one in prices and increasing in $\bar{u}$.
- (c) Verify Shephard's lemma for good x : show $\partial e / \partial p_x = h_x$.
- (d) In two or three sentences, explain how Hicksian demand differs from Marshallian demand, and why — for a normal good — the Hicksian (compensated) demand curve is *steeper* than the Marshallian one.

Question 2 — Slutsky decomposition, with numbers (both directions)

The same consumer, $u(x, y) = x^{0.5}y^{0.5}$, has income $m = 80$ and initially faces $p_x = 2$, $p_y = 2$. The price of x falls to $p'_x = 1$.

- (a) Find the initial bundle (x_0, y_0) and the new bundle (x_1, y_1) , and hence the *total* change in x .
- (b) Using your Hicksian demands from Q1, find the compensated bundle at the new prices holding the *original* utility, and the income e needed to buy it.
- (c) Decompose Δx into substitution and income effects. Report both numbers and check they sum to the total.
- (d) Is x normal or inferior here, and how do you know from your two effects? In two sentences, state what would have to be true of the income effect for this good to be Giffen, and why Cobb–Douglas can never produce a Giffen good.
- (e) *Mirror case.* Now suppose instead the price *rises* from $p_x = 1$ to $p_x = 2$ (with $m = 80$, $p_y = 2$). Repeat the decomposition and confirm that both effects are now *negative* for this normal good.

Question 3 — Elasticity, functional form, and revenue

- (a) For the constant-elasticity demand $Q = Ap^{-\eta}$ ($A, \eta > 0$), show the own-price elasticity is $-\eta$ everywhere, and find the elasticity of *total revenue* with respect to price. State the condition on η under which a price rise raises revenue. What is η for Cobb–Douglas demand $x^* = \alpha m/p_x$, and what does that imply for spending on x ?
- (b) For the linear demand $Q = 150 - 3p$, find: (i) the unit-elastic price, (ii) the revenue-maximising price, (iii) the price elasticity at $p = 10$ and at $p = 40$, and (iv) classify each as elastic/inelastic.
- (c) A manager says “our demand curve is a straight line, so our elasticity is constant.” Using your Q3(b) results, explain in 2–3 sentences exactly why this is wrong and what *is* constant.

Question 4 — A real elasticity, found and interpreted

- (a) Locate a published estimate of the price elasticity of demand for a specific good (e.g. petrol, cigarettes, electricity, a named foodstuff) from a credible source — a government report, regulator, or peer-reviewed paper. Report the number, the good, the country/period, and full citation.
- (b) State whether short-run and long-run estimates differ for your good, and explain *why* the time horizon changes elasticity (give the mechanism, not just the direction).
- (c) Suppose a government imposes a 15% price-raising tax on your good. Using your cited elasticity, estimate the percentage change in quantity demanded and comment on who bears most of the burden.
- (d) Name one reason your back-of-envelope estimate in (c) could be badly wrong (e.g. the elasticity is local, non-linearity, the tax is large relative to the data range).

Question 5 — Cross-price and market definition

Two products A and B have estimated cross-price elasticity $\varepsilon_{AB} = +1.8$.

- (a) Interpret the sign and magnitude. Are A and B substitutes or complements, and strongly so?
- (b) A competition authority must decide whether A and B are “in the same relevant market.” Explain how ε_{AB} informs that decision and what threshold reasoning a regulator might apply.
- (c) Give a concrete real example of two products you would expect to have a large positive cross-price elasticity, and one pair with a large *negative* one. Justify each in one sentence.

Solutions

Question 1

(a) Tangency: $MRS = \frac{MU_x}{MU_y} = \frac{0.5x^{-0.5}y^{0.5}}{0.5x^{0.5}y^{-0.5}} = \frac{y}{x} = \frac{p_x}{p_y} \Rightarrow y = \frac{p_x}{p_y}x$. Impose $x^{0.5}y^{0.5} = \bar{u}$:
 $x^{0.5}\left(\frac{p_x}{p_y}x\right)^{0.5} = x\left(\frac{p_x}{p_y}\right)^{0.5} = \bar{u}$, so

$$h_x = \bar{u} \left(\frac{p_y}{p_x} \right)^{1/2}, \quad h_y = \bar{u} \left(\frac{p_x}{p_y} \right)^{1/2}.$$

Both compensated demands fall in their own price — the compensated demand curve always slopes down.

(b) $e = p_x h_x + p_y h_y = p_x \bar{u} \left(\frac{p_y}{p_x} \right)^{1/2} + p_y \bar{u} \left(\frac{p_x}{p_y} \right)^{1/2} = \bar{u} \sqrt{p_x p_y} + \bar{u} \sqrt{p_x p_y} = 2\bar{u} \sqrt{p_x p_y}$. Scaling both prices by t : $e(tp_x, tp_y, \bar{u}) = 2\bar{u} \sqrt{t^2 p_x p_y} = t e$ — homogeneous of degree 1. And $\partial e / \partial \bar{u} = 2\sqrt{p_x p_y} > 0$ — increasing in the utility target.

(c) $\frac{\partial e}{\partial p_x} = \frac{\partial}{\partial p_x} (2\bar{u} p_x^{1/2} p_y^{1/2}) = 2\bar{u} \cdot \frac{1}{2} p_x^{-1/2} p_y^{1/2} = \bar{u} \left(\frac{p_y}{p_x} \right)^{1/2} = h_x$. ✓

(d) Marshallian demand holds *money income* m fixed, so a price change moves the consumer to a new indifference curve — it carries both a substitution and an income effect. Hicksian demand holds *utility* $\bar{u}$ fixed (the consumer is compensated), so it is the pure substitution response. For a normal good the income effect reinforces the substitution effect, so the Marshallian curve moves more per unit price change — i.e. it is *flatter*, and the Hicksian is *steeper*.

Question 2

(a) Cobb–Douglas with equal exponents spends half of income on each good, so $x^* = 0.5m/p_x$, $y^* = 0.5m/p_y$. Initial ($p_x = p_y = 2$, $m = 80$): $x_0 = \frac{0.5 \cdot 80}{2} = 20$, $y_0 = 20$. After $p'_x = 1$: $x_1 = \frac{0.5 \cdot 80}{1} = 40$, $y_1 = 20$. Total change $\Delta x = +20$.

(b) Original utility $\bar{u} = \sqrt{20 \cdot 20} = 20$. Compensated (Hicksian) bundle at new prices (1, 2) using Q1: $h_x = 20\sqrt{2/1} = 20\sqrt{2} \approx 28.28$, $h_y = 20\sqrt{1/2} = 10\sqrt{2} \approx 14.14$. Required income $e = 2\bar{u}\sqrt{p_x p_y} = 2(20)\sqrt{1 \cdot 2} = 40\sqrt{2} \approx 56.57$ (equivalently $1(28.28) + 2(14.14)$).

(c) Substitution effect = $x_c - x_0 = 28.28 - 20 = +8.28$. Income effect = $x_1 - x_c = 40 - 28.28 = +11.72$. They sum to $+20 = \Delta x$. ✓

(d) x is **normal**: the income effect is positive (the price fall raises real income, raising x). For a Giffen good the income effect would have to be negative *and* larger in magnitude than the substitution effect. Cobb–Douglas demand $x = \alpha m/p_x$ is strictly increasing in m (always normal, never inferior), so it can never be Giffen.

(e) Now the start is $p_x = 1$: $x_0 = 40$, $y_0 = 20$, $\bar{u} = \sqrt{40 \cdot 20} = \sqrt{800} \approx 28.28$. New prices (2, 2): $x_1 = \frac{0.5 \cdot 80}{2} = 20$; total $\Delta x = -20$. Compensated at $p_x = 2$ holding $\bar{u} = 28.28$: $h_x = 28.28\sqrt{2/2} = 28.28$. Substitution = $28.28 - 40 = -11.72$; income = $20 - 28.28 = -8.28$; sum = -20 . Both negative — a dearer x is substituted away (SE < 0 always) and the loss of real income cuts x further because it is normal (IE < 0).

Question 3

(a) $Q = Ap^{-\eta}$: $\varepsilon = \frac{dQ}{dp} \frac{p}{Q} = A(-\eta)p^{-\eta-1} \frac{p}{Ap^{-\eta}} = -\eta$ at every price. Revenue $R = pQ = Ap^{1-\eta}$, so the elasticity of revenue w.r.t. price is $1 - \eta$. A price rise raises revenue iff $1 - \eta > 0$, i.e. $\eta < 1$ (inelastic demand). Cobb–Douglas $x^* = \alpha m/p_x$ is the case $\eta = 1$: own-price elasticity $\equiv -1$, so spending $p_x x^* = \alpha m$ on x is *constant* at every price.

(b) $Q = 150 - 3p$, $\varepsilon = \frac{-3p}{150 - 3p}$. (i) Unit-elastic: $-3p = -(150 - 3p) \Rightarrow p = 25$. (ii) Revenue $R = 150p - 3p^2$, $R' = 150 - 6p = 0 \Rightarrow p = 25$ (same point). (iii) $p = 10$: $\varepsilon = \frac{-30}{120} = -0.25$; $p = 40$: $\varepsilon = \frac{-120}{30} = -4$. (iv) $p = 10$ *inelastic*; $p = 40$ *elastic*.

(c) The manager confuses slope with elasticity. Along a linear demand the *slope* $dQ/dp = -3$ is constant, but elasticity = slope $\times (p/Q)$ varies with p/Q — from 0 near $p = 0$ to $-\infty$ as $Q \rightarrow 0$ (see the -0.25 vs -4 above). Constant slope, not constant elasticity.

Question 4 (model answer / marking notes)

Data-dependent. (a) reports a cited published own-price elasticity (good, country/period, source). (b) explains why long-run $|\varepsilon|$ typically exceeds short-run: more substitution margins open up over time (capital stock, habits, technology adjust) — give the *mechanism*, e.g. drivers eventually buy more efficient cars or switch modes. (c) $\% \Delta Q \approx \varepsilon \times 15\%$; the burden falls more on the side that is relatively *inelastic* (if $|\varepsilon| < 1$ consumers bear most of it). (d) a valid caveat: the estimate is local/linearised, a 15% change may exceed the data range, the cited ε may be for a different period/market, or non-linearity in demand.

Question 5

(a) $\varepsilon_{AB} = +1.8 > 0 \Rightarrow A$ and B are **substitutes**; magnitude > 1 means *strong* substitutability (a 1% rise in B 's price raises demand for A by 1.8%).

(b) A high positive cross-price elasticity is evidence the two products constrain each other's pricing, so a regulator is more likely to place them in the *same relevant market*. This is the logic of the SSNIP/hypothetical-monopolist test: if a small but significant non-transitory price rise on A would drive enough switching to B to make it unprofitable, B belongs in the market. Authorities apply rough thresholds (notably elasticities well above zero) rather than a single bright line.

(c) *Model answer*. Large positive cross-price elasticity: two brands of the same product (e.g. competing supermarket colas), because buyers switch readily on price. Large negative: a durable and its consumable (e.g. game console and its games), because a dearer console depresses demand for its complementary titles.